Random Processes

Chapter 2: Practice problems (Lectures 6–7(b))

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Scope. Periodicity, the limit theorems, coupling and total variation. Result numbers refer to the Chapter 2 notes (e.g. Theorem 6.7 = limit theorems, Lemma 7.8 = eventual positivity, Theorem 7.22 = maximal coupling). A $\star$ on a problem or a part means it is harder than the Fall 2026 practice midterm (proofs, or computations beyond plugging into a theorem); unmarked problems are at about exam level. Solutions start on a new page after the problems.

Problems

Problem 1 (Four-state chain).
On $S = \{1, 2, 3, 4\}$ let

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

(a) Show the chain is irreducible and find the period of each state.

(b) Find the stationary distribution π, and compute $\mathbb{E}_1 T_1$ and $\mathbb{E}_4 T_4$ two ways: from π, and directly from the return paths.

(c) Find $\lim_{n\to\infty} P^n(4, 1)$. Which part of Theorem 6.7 justifies this?

(d) You earn \$7 in every time step the chain is at state 4 and nothing otherwise. What is the long-run average earning per step, almost surely?

(e) $\star$ Determine $I_1 = \{n \geq 1 : P^n(1, 1) > 0\}$ and the smallest n_1 with $P^n(1, 1) > 0$ for all $n \geq n_1$.

(f) $\star$ The smallest n with $P^n > 0$ entrywise is $n = 10$. Explain why P^9 still has a zero entry.

Problem 2 (Periodic chain on a cycle).
Let (X_n) be simple random walk on the cycle $\mathbb{Z}_6 = \{0, 1, ..., 5\}$: $p(x, x \pm 1) = \frac{1}{2} \pmod 6$.

(a) Find the period and the stationary distribution. Does $P^n(0, 0)$ converge?

(b) $\star$ Compute $\lim_n P^{2n}(0, y)$ and $\lim_n P^{2n+1}(0, y)$ for every y. (Hint: look at P^2 restricted to the even states.)

(c) Compute $\lim_n \frac{1}{n} \sum_{j=1}^{n} P^j(0, 0)$ and the almost-sure long-run fraction of time spent at 0. Which hypotheses of Theorem 6.7 are you using?

Problem 3 ⋆ (Cyclic classes).
Let P be irreducible with period d. Fix $x, y \in S$. Show that all n with $P^n(x,y) > 0$ are congruent modulo d.

Problem 4 (Exact convergence rate for two states).
Let $P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $a, b \in [0,1]$, $a+b>0$, and put $\lambda = 1-a-b$.
(a) Show that $P^n(1,1) = \frac{b}{a+b} + \frac{a}{a+b}\lambda^n$.
(b) Compute $\|P^n(1,\cdot) - \pi\|_{\mathrm{TV}}$ exactly.
(c) For which (a,b) does $P^n(1,\cdot)$ fail to converge? Relate this to periodicity.

Problem 5 ⋆ (Total variation is non-increasing).
Let μ, ν be probability distributions on a countable S and P a transition matrix.
(a) Show $\|\mu P - \nu P\|_{\mathrm{TV}} \le \|\mu - \nu\|_{\mathrm{TV}}$, once using the formula $\frac{1}{2}\sum|\cdot|$ and once using a coupling.
(b) Deduce that if π is stationary, then $n \mapsto \|P^n(x,\cdot) - \pi\|_{\mathrm{TV}}$ is non-increasing.

Problem 6 ⋆ (Limiting $\Longrightarrow$ unique stationary, and any initial law).
(a) Suppose q is a limiting distribution for P on countable S. Show that every stationary distribution π equals q. (Use DCT on $\pi(y) = \sum_x \pi(x)P^n(x,y)$.)
(b) Suppose P is irreducible, aperiodic and has a stationary π. Show that $\|\mu P^n - \pi\|_{\mathrm{TV}} \to 0$ for every initial distribution μ.

Problem 7 ⋆ (Eventual positivity with a self-loop).
Let P be irreducible on a finite S with $|S| = k$, and suppose $p(z,z) > 0$ for some z. Show $P^n(x,y) > 0$ for all x, y and all $n \ge 2k-2$. Compare with Problem 1(f): why doesn't this bound apply there?

Problem 8 (Aperiodicity in the coupling proof).
Let $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
(a) Write down the product chain $Q((u,v),(u',v')) = p(u,u')p(v,v')$ on $\{1,2\}^2$ and show it is not irreducible.
(b) Run two independent copies from $(X_0, Y_0) = (1,2)$. Show the meeting time is $\tau = \infty$ almost surely.
(c) Point to the exact step in the proof of Theorem 6.7(1) that breaks.

Problem 9 (TV and a maximal coupling).
Let $1 \le m < n$, $\mu = \mathrm{Unif}\{1,\dots,m\}$ and $\nu = \mathrm{Unif}\{1,\dots,n\}$.
(a) Compute $\|\mu - \nu\|_{\mathrm{TV}}$.
(b) ⋆ Construct a coupling (U,V) with $\mathbb{P}(U \ne V) = \|\mu - \nu\|_{\mathrm{TV}}$.

Problem 10 (Maximal coupling by the formula).
Let $p = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ and $q = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ on $\{1,2,3\}$. Compute $\|p - q\|_{\mathrm{TV}}$ and write the 3×3 joint table γ of the maximal coupling from Theorem 7.22. Check both marginals.

Problem 11 ⋆ (Poisson approximation (Le Cam)).
(a) Show $\|\mathrm{Ber}(p) - \mathrm{Pois}(p)\|_{\mathrm{TV}} = p(1 - e^{-p}) \le p^2$ for $p \in [0, 1]$.
(b) Using (a) and a coupling, show $\|\mathrm{Bin}(n, p) - \mathrm{Pois}(np)\|_{\mathrm{TV}} \le np^2$.
(c) Give the bound for $n = 100$, $p = 0.02$.

Problem 12 (Biased walk on $\mathbb{Z}$).
Let $p(x, x+1) = p$ and $p(x, x-1) = q = 1 - p$ on $\mathbb{Z}$, with $p \in (0, 1)$.
(a) Find $P^{2n}(0, 0)$ and its asymptotics using $\binom{2n}{n} \sim 4^n/\sqrt{\pi n}$.
(b) Decide recurrence or transience for each p.
(c) ⋆ Is there a stationary distribution when $p = \frac{1}{2}$? (Use Theorem 6.7(3) and translation invariance.)

Problem 13 (Random-to-top, numbers).
Take $m = 52$ cards and $\varepsilon = 0.01$; $H_{52} \approx 4.538$.
(a) How many steps n guarantee $\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \le \varepsilon$ via the Markov-inequality bound mH_m/n? Via the union bound $me^{-n/m}$?
(b) (⋆ Lower bound.) Show that for $n \le m - 2$, with $j = m - n$,

$$\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \ge 1 - \frac{1}{j!}.$$

(Hint: which cards can be at the bottom after n steps, and in what order?) What does this give for $n = 50$ and $n = 40$?

Problem 14 (True or false; justify briefly).
(a) If P is irreducible on a finite set, then P^n converges entrywise.
(b) A stationary distribution is always a limiting distribution.
(c) If $x \leftrightarrow y$ then $d(x) = d(y)$.
(d) If π is stationary and $\pi(y) = 0$, then y is transient.
(e) For a finite irreducible chain, $\frac{1}{n}\sum_{j=1}^{n} P^j(x, y)$ converges.
(f) $\|\mu - \nu\|_{\mathrm{TV}} = \sum_z |\mu(z) - \nu(z)|$.
(g) An irreducible chain with $p(x, x) > 0$ for at least one x is aperiodic.
(h) If $U \sim \mu$, $V \sim \nu$ and $\mathbb{P}(U \ne V) = 0.1$, then $\|\mu - \nu\|_{\mathrm{TV}} = 0.1$.
(i) The ergodic theorem (Theorem 6.7(4)) requires aperiodicity.

Solutions

Solution 1.
(a) $1 \to 2 \to 3 \to 1$ and $3 \to 4 \to 1$, so every state reaches every other: the chain is irreducible. Return loops at 1 have lengths 3 ($1 \to 2 \to 3 \to 1$) and 4 ($1 \to 2 \to 3 \to 4 \to 1$), and $\gcd(3,4) = 1$. Period is a class property (Lemma 6.4), so every state has period 1.

(b) $\pi P = \pi$ reads $\pi_1 = \frac{1}{2}\pi_3 + \pi_4$, $\pi_2 = \pi_1$, $\pi_3 = \pi_2$, $\pi_4 = \frac{1}{2}\pi_3$. Take $\pi_1 = \pi_2 = \pi_3 = 2$, $\pi_4 = 1$ (check: $\pi_1 = 1 + 1$) and normalise:

$$\pi = \left(\frac{2}{7}, \frac{2}{7}, \frac{2}{7}, \frac{1}{7}\right), \quad \mathbb{E}_1 T_1 = \frac{1}{\pi_1} = \frac{7}{2}, \quad \mathbb{E}_4 T_4 = \frac{1}{\pi_4} = 7.$$

Directly: from 1 the return takes 3 or 4 steps with probability $\frac{1}{2}$ each, so $\mathbb{E}_1 T_1 = \frac{7}{2}$. From 4: go $4 \to 1 \to 2 \to 3$ (3 steps). At 3, either go to 4 (one more step), or go to 1 and repeat the loop $3 \to 1 \to 2 \to 3$. The number G of such loops is geometric on $\{0, 1, ...\}$ with success probability $\frac{1}{2}$, so $\mathbb{E}G = 1$ and $\mathbb{E}_4 T_4 = 3 + 1 + 3\mathbb{E}G = 7$.

(c) Irreducible, aperiodic and π exists, so Theorem 6.7(1) gives $P^n(4,1) \to \pi_1 = \frac{2}{7}$.

(d) Theorem 6.7(4) with $f = 7 \cdot \mathbf{1}_{\{4\}}$: the average tends to $\sum_z f(z)\pi(z) = 7 \cdot \frac{1}{7} = 1$ dollar per step.

(e) Every return to 1 is a concatenation of 3-loops and 4-loops, so

$$I_1 = \{3a + 4b : a, b \geq 0, a + b \geq 1\} = \{3, 4, 6, 7, 8, 9, ...\}.$$

The integers $1, 2, 5$ are not of this form, and every $n \geq 6$ is ($6 = 3 + 3$, $7 = 3 + 4$, $8 = 4 + 4$, then add 3's). So $n_1 = 6$.

(f) A return to 4 is the path $4 \to 1 \to 2 \to 3 \to 4$ (length 4) with some number of 3- or 4-loops through 1 inserted, so $I_4 = \{4 + 3a + 4b\} = \{4, 7, 8, 10, 11, ...\}$. Since $9 - 4 = 5$ is not of the form $3a + 4b$, $P^9(4,4) = 0$.

Solution 2.
(a) Each step changes parity, and $0 \to 1 \to 0$ has length 2, so $d = 2$. P is doubly stochastic, so $\pi \equiv \frac{1}{6}$ (unique by irreducibility). $P^n(0,0) = 0$ for odd n, but by (b) $P^{2n}(0,0) \to \frac{1}{3}$, so $P^n(0,0)$ does not converge.

(b) By parity (or Problem 3), $P^{2n}(0,y) = 0$ for odd y and $P^{2n+1}(0,y) = 0$ for even y. The two-step chain on $E = \{0, 2, 4\}$ has $P^2(x,x) = \frac{1}{2}$ and $P^2(x, x \pm 2) = \frac{1}{4}$: it is irreducible, aperiodic (self-loop) and doubly stochastic. So its stationary law is uniform on E, and Theorem 6.7(1) gives $P^{2n}(0,y) \to \frac{1}{3}$ for $y \in E$. Then $P^{2n+1}(0,y) = \sum_z p(0,z)P^{2n}(z,y) \to \frac{1}{3}$ for odd y, by the same argument on the odd class. So

$$\lim_n P^{2n}(0,\cdot) = \left(\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0\right), \quad \lim_n P^{2n+1}(0,\cdot) = \left(0, \frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}\right).$$

(c) The Cesàro limit is the average of $\frac{1}{3}$ and 0, namely $\frac{1}{6} = \pi(0)$, consistent with Theorem 6.7(5). The almost-sure fraction of time at 0 is $1/\mathbb{E}_0 T_0 = \pi(0) = \frac{1}{6}$ by Theorem 6.7(2),(3). These parts need only I + R or I + S, not aperiodicity.

Solution 3.
Choose $m \geq 1$ with $P^m(y,x) > 0$. If $P^{n_1}(x,y) > 0$ and $P^{n_2}(x,y) > 0$, then by Chapman–Kolmogorov

$$P^{n_i+m}(x,x) \geq P^{n_i}(x,y)P^m(y,x) > 0 \quad (i = 1,2),$$

so $n_1 + m, n_2 + m \in I_x$. Both are divisible by $d = d(x)$, hence $d \mid n_1 - n_2$.

Solution 4.
(a) Let $u_n = P^n(1,1)$. Then $u_{n+1} = u_n(1-a) + (1-u_n)b = b + \lambda u_n$ with $u_0 = 1$. The fixed point is $u^* = \frac{b}{a+b}$, and $u_n - u^* = \lambda^n(1-u^*) = \frac{a}{a+b}\lambda^n$.
(b) $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$ and $P^n(1,\cdot) - \pi = \frac{a}{a+b}\lambda^n(1,-1)$, so

$$\|P^n(1,\cdot) - \pi\|_{\mathrm{TV}} = \frac{1}{2} \cdot 2 \cdot \frac{a}{a+b}|\lambda|^n = \frac{a}{a+b}|1-a-b|^n.$$

(c) Since $a + b > 0$, $|\lambda| < 1$ unless $\lambda = -1$, i.e. $a = b = 1$. That is the flip chain, which is irreducible with period 2; $P^n(1,1)$ alternates between 0 and 1. (If $a = b = 0$, then $P = I$, which is reducible and every distribution is stationary.)

Solution 5.
(a) *Formula.* By Tonelli,

$$\|\mu P - \nu P\|_{\mathrm{TV}} = \frac{1}{2}\sum_y \left|\sum_x (\mu(x) - \nu(x))p(x,y)\right| \leq \frac{1}{2}\sum_x |\mu(x) - \nu(x)| \sum_y p(x,y) = \|\mu - \nu\|_{\mathrm{TV}}.$$

Coupling. Take a maximal coupling (X_0, Y_0) of μ, ν (Theorem 7.22). If $X_0 = Y_0$, take one step together; otherwise step independently. Then $X_1 \sim \mu P$, $Y_1 \sim \nu P$, and $\{X_1 \neq Y_1\} \subseteq \{X_0 \neq Y_0\}$. So

$$\|\mu P - \nu P\|_{\mathrm{TV}} \leq \mathbb{P}(X_1 \neq Y_1) \leq \mathbb{P}(X_0 \neq Y_0) = \|\mu - \nu\|_{\mathrm{TV}}.$$

(b) Apply (a) with $\mu = P^n(x,\cdot)$ and $\nu = \pi$, using $\pi P = \pi$:

$$\left\|P^{n+1}(x,\cdot) - \pi\right\|_{\mathrm{TV}} = \|P^n(x,\cdot)P - \pi P\|_{\mathrm{TV}} \leq \|P^n(x,\cdot) - \pi\|_{\mathrm{TV}}.$$

Solution 6.
(a) Fix y. For every n, $\pi(y) = (\pi P^n)(y) = \sum_x \pi(x)P^n(x,y)$. The summand tends to $\pi(x)q(y)$ and is dominated by $\pi(x)$, with $\sum_x \pi(x) = 1 < \infty$. DCT (counting measure on S) gives $\pi(y) = \sum_x \pi(x)q(y) = q(y)$.
(b) Since $\pi = \sum_x \mu(x)\pi$,

$$\|\mu P^n - \pi\|_{\mathrm{TV}} = \frac{1}{2}\sum_y \left|\sum_x \mu(x)(P^n(x,y) - \pi(y))\right| \leq \sum_x \mu(x)\|P^n(x,\cdot) - \pi\|_{\mathrm{TV}}.$$

Each term tends to 0 by Theorem 6.7(1) and is dominated by $\mu(x)$; DCT gives the limit 0.

Solution 7.
Fix x, y. By irreducibility there are paths $x \to z$ of length $\ell_1 \le k-1$ and $z \to y$ of length $\ell_2 \le k-1$: a shortest path visits distinct states, and the length is 0 if the endpoints coincide. For $n \ge 2k-2$, put $s = n - \ell_1 - \ell_2 \ge 0$ and insert s self-loops at z:

$$P^n(x,y) \ge P^{\ell_1}(x,z) \cdot p(z,z)^s \cdot P^{\ell_2}(z,y) > 0.$$

In Problem 1 no state has a self-loop, so the argument does not apply; there the threshold is $10 > 2 \cdot 4 - 2 = 6$.

Solution 8.
(a) Q moves $(1,1) \leftrightarrow (2,2)$ and $(1,2) \leftrightarrow (2,1)$ deterministically. There are two closed classes, the diagonal $\{(1,1),(2,2)\}$ and the off-diagonal $\{(1,2),(2,1)\}$, so Q is not irreducible.

(b) From $(1,2)$ the pair alternates $(1,2),(2,1),(1,2),...$ and never reaches the diagonal, so $\tau = \infty$.

(c) Step 1 of the proof uses aperiodicity (Lemma 7.8 and Corollary 7.9) to show that Q is irreducible. That is what makes the recurrent state (a,a) reachable from every starting pair, giving $\mathbb{P}(\tau < \infty) = 1$. Here that step fails. Consistently, $\|P^n(1,\cdot) - \pi\|_{\mathrm{TV}} = \frac{1}{2}$ for all n.

Solution 9.
(a) $\sum_z \min(\mu(z), \nu(z)) = m \cdot \frac{1}{n}$, so $\|\mu - \nu\|_{\mathrm{TV}} = 1 - \frac{m}{n}$.

(b) Let $V \sim \mathrm{Unif}\{1,...,n\}$. If $V \le m$ put $U = V$; otherwise draw $U \sim \mathrm{Unif}\{1,...,m\}$ independently. Then for $u \le m$,

$$\mathbb{P}(U = u) = \frac{1}{n} + \frac{n-m}{n} \cdot \frac{1}{m} = \frac{1}{m},$$

so $U \sim \mu$, and $\mathbb{P}(U \ne V) = \mathbb{P}(V > m) = 1 - \frac{m}{n}$. By the coupling inequality (Lemma 7.5) this is optimal.

Solution 10.
$r = \min(p,q) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{6}\right)$, so $\sum r = \frac{5}{6}$ and $d = \|p - q\|_{\mathrm{TV}} = \frac{1}{6}$. The excesses are $a = p - r = \left(\frac{1}{6}, 0, 0\right)$ and $b = q - r = \left(0, 0, \frac{1}{6}\right)$. Off the diagonal $\gamma(x,y) = a_x b_y / d$, which is nonzero only at $(1,3)$: $\gamma(1,3) = \frac{\frac{1}{6} \cdot \frac{1}{6}}{\frac{1}{6}} = \frac{1}{6}$. So

$$\gamma = \begin{pmatrix} \frac{1}{3} & 0 & \frac{1}{6} \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{6} \end{pmatrix}.$$

The row sums are $\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = p$, the column sums are $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = q$, and $\mathbb{P}(U \ne V) = \frac{1}{6} = d$.

Solution 11.
(a) Compute the overlaps. At 0: $\min(1-p, e^{-p}) = 1 - p$, since $e^{-p} \ge 1 - p$. At 1: $\min(p, pe^{-p}) = pe^{-p}$. At $k \ge 2$, $\mathrm{Ber}(p)$ has no mass. Hence

$$\|\mathrm{Ber}(p) - \mathrm{Pois}(p)\|_{\mathrm{TV}} = 1 - (1-p) - pe^{-p} = p(1 - e^{-p}) \le p \cdot p,$$

using $1 - e^{-p} \le p$.

(b) Take independent pairs $(U_i, V_i)_{i=1}^n$, each a maximal coupling of $\mathrm{Ber}(p)$ and $\mathrm{Pois}(p)$. Then $\sum_i U_i \sim \mathrm{Bin}(n,p)$ and $\sum_i V_i \sim \mathrm{Pois}(np)$, since independent Poissons add. The coupling inequality and a union bound give

$$\|\mathrm{Bin}(n,p) - \mathrm{Pois}(np)\|_{\mathrm{TV}} \le \mathbb{P}\left(\sum_i U_i \neq \sum_i V_i\right) \le \sum_i \mathbb{P}(U_i \neq V_i) \le np^2.$$

(c) $100 \cdot 0.02^2 = 0.04$.

Solution 12.

(a) A return in $2n$ steps needs n up-steps and n down-steps, so

$$P^{2n}(0,0) = \binom{2n}{n}(pq)^n \sim \frac{(4pq)^n}{\sqrt{\pi n}},$$

and odd-step returns are impossible (period 2).

(b) $4pq = 1 - (p-q)^2 \le 1$, with equality iff $p = \frac{1}{2}$. If $p \neq \frac{1}{2}$, the terms decay geometrically, so $\sum_n P^n(0,0) < \infty$ and the walk is transient. If $p = \frac{1}{2}$, the terms are $\sim 1/\sqrt{\pi n}$, whose sum diverges, so the walk is recurrent. Irreducibility extends this to every state.

(c) No. A stationary π would give $\pi(x) = 1/\mathbb{E}_x T_x$ for all x (Theorem 6.7(3)). Translation invariance makes $\mathbb{E}_x T_x$ the same for every x, so π would be constant on $\mathbb{Z}$, which cannot sum to 1. (In Lecture 9 terms, the walk is null recurrent.)

Solution 13.

(a) Markov bound: $mH_m/n \le 0.01$ needs $n \ge 52 \cdot 4.538/0.01 \approx 23\,598$. Union bound: $me^{-n/m} \le 0.01$ needs $n \ge m(\log m + \log 100) \approx 52(3.951 + 4.605) \approx 444.9$, so $n = 445$ suffices.

(b) After n steps at most n distinct cards have been selected, so at least $j = m - n$ cards are unselected. Unselected cards sit at the bottom of the deck in their original relative order. Let A be the event that the bottom j cards appear in the same relative order as in x. Then $P^n(x, A) = 1$. Under π, the order of the bottom j cards is uniform among the $j!$ orders, so $\pi(A) = 1/j!$. Hence

$$\|P^n(x,\cdot) - \pi\|_{\mathrm{TV}} \ge P^n(x,A) - \pi(A) = 1 - \frac{1}{j!}.$$

For $n = 50$ ($j = 2$) this is at least $\frac{1}{2}$. For $n = 40$ ($j = 12$) it is at least $1 - 1/12! \approx 1 - 2.1 \times 10^{-9}$. So roughly m steps are necessary, while $m \log m$ steps suffice.

Solution 14.

(a) False: the flip chain has $P^{2n} = I$ and $P^{2n+1} = P$.

(b) False: in the same example $\pi = \left(\frac{1}{2}, \frac{1}{2}\right)$ is stationary but there is no limit.

(c) True: Lemma 6.4.

(d) False: take $P = I$ on $\{1,2\}$ and $\pi = \delta_1$. Then $\pi(2) = 0$, but 2 is absorbing, hence recurrent. (For an irreducible chain the statement is vacuous, because then $\pi > 0$.)

(e) True: Theorem 6.7(5); a finite irreducible chain has I + S.

(f) False: $\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2}\sum_z |\mu(z) - \nu(z)|$.

(g) True: $1 \in I_x$ gives $d(x) = 1$, and period is a class property.

(h) False: the coupling inequality only gives $\|\mu - \nu\|_{\mathrm{TV}} \leq 0.1$, with equality only for a maximal coupling. For example, $\mu = \nu$ can be coupled to disagree.

(i) False: it needs only I + S.